Mental Health Prediction Model

Group 3

By

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4. Introduction

This project focuses on studying the relationship between workplace conditions and employees mental health to understand the factors that influence emotional well-being and performance.
We will build models that predict possible mental health challenges and identify individuals who may need treatment or psychological support .
The findings of this project will help organizations develop strategies that support mental stability ,reduce work related stress and enhance overall productivity [8].

5. Project Approach

Methodology:

This project follows the **CRISP-DM** methodology, which includes six main phases:

1. **Business Understanding:** define project goals related to workplace mental health.
2. **Data Understanding:** explore and analyze data to identify key factors.
3. **Data Preparation:** clean and organize data for modeling.
4. **Modeling:** apply machine learning models to predict mental health challenges.
5. **Evaluation:** assess model performance and accuracy.
6. **Deployment:** this phase is considered as *future work*, where the developed model can be implemented and expanded to real workplace environments.[9]

2.1 Step 1 – Data Preprocessing

before applying any model, we performed several preprocessing steps to prepare and clean the dataset.
First, we uploaded the dataset into Google Colab and used functions such as head(), info(), and describe() to understand its structure and identify possible issues like missing values or outliers.

We found that the columns work_interfere and self_employed contained missing values, which were handled later to keep the data complete. Outliers in the Age column were detected using the IQR method and visualized with a boxplot, then cleaned to improve data reliability.

Next, we separated numerical and categorical features to apply appropriate handling numeric values were filled with the mean and categorical ones with the mode.
After that, Label Encoding was applied to convert categorical data into numerical form, and StandardScale**r** was used to scale the Age column to ensure better model performance.

These preprocessing steps helped us make the dataset clean, consistent, and ready for modeling. [2]

30]
0s

```
missing_values = copydata.isnull().sum()
missing_percentage = (missing_values / len(copydata)) * 100
missing_report = pd.DataFrame({'Missing Values': missing_values, 'Percentage': missing_percentage})
```

16]
0s

```
print(" Missing Values Report:\n", missing_report[missing_report['Missing Values'] > 0])
```

```
 Missing Values Report:
                 Missing Values  Percentage
self_employed               18    1.429706
work_interfere             264   20.969023
```

17]
0s

```
sns.boxplot(x=copydata['Age'])
plt.title('Age Outliers Visualization')
plt.show()
```

Age Outliers Visualization

2.2 Step 2 – Models Training

In this step, we will apply four supervised machine learning classifiers to predict whether employees are likely to seek mental health treatment based on workplace and personal factors.

The selected classifiers will be Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM). These algorithms are chosen because they are interpretable, efficient, and capable of handling both numerical and categorical data.

All models will be implemented using Python with the Scikit-learn library, which provides robust tools for model training, testing, and evaluation. We will divide the dataset into training and testing subsets using the train_test_split() function to ensure fair evaluation and to minimize overfitting.

During the training process, each classifier will be trained on 80% of the dataset and tested on the remaining 20%. All models will be trained on the same preprocessed and normalized data to ensure consistency in comparison.

Each algorithm will learn patterns and relationships from the input features to predict the target variable (*treatment*). Afterward, we will compare their performance to identify the most suitable model for this prediction task.

Step 2 - Models Training

```
X = copydata.drop(columns=['treatment'])
y = copydata['treatment']
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

2.3 Step 3 – Comparative Analysis

In this step, we will compare the performance of four classification models Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) to determine which one provides the most accurate and consistent predictions. The goal is to evaluate how effectively each model can predict whether an employee is likely to seek mental health treatment based on workplace and personal factors. To achieve this, we will use several performance metrics, each offering a different perspective on model performance:

- **Accuracy:**
  We will measure how often each model makes correct predictions overall. It represents the ratio of correctly classified samples (both treatment and non-treatment) to the total number of samples.
- **Precision:**
  This metric will show how many of the employees predicted as "seeking treatment" were actually correct. A high precision value indicates that the model makes fewer false positive predictions.
- **Recall:**
  We will evaluate how well the model identifies employees who truly need treatment. A higher recall value means the model captures most of the actual positive cases, minimizing false negatives.
- **F1-Score:**
  This score will be calculated to combine both Precision and Recall into a single value. It provides a balanced measure when the dataset is not equally distributed between classes which is important in our case, since the number of employees seeking treatment may differ from those who do not.
- **ConfusionMatrix:**
  We will use this table to visualize model performance by showing the number of correct and incorrect predictions. It will help us understand how many employees were correctly classified as seeking or not seeking treatment, and where the model made errors.

By analyzing these metrics together, we will be able to assess the strengths and weaknesses of each classifier and identify the model that performs best for predicting mental health treatment decisions.

2.4 Step 4 – Best Performing Models

In this step, the performance of all classifiers is analyzed to determine which model provides the most accurate and balanced predictions. The comparison focuses on accuracy, precision, recall, and F1-score to ensure a fair evaluation.

A model is considered effective when it achieves high accuracy while maintaining a good balance between precision and recall, allowing it to correctly identify employees who truly need treatment. For this project, models with an accuracy above 80% and an F1-score around or higher than 0.70 are considered reliable.[7]

The confusion matrix is also reviewed to ensure that false negatives remain low, as missing actual treatment cases is critical in mental health prediction. Overall, the best performing model is the one that shows consistent and stable results across all metrics.

6. Project Requirements

3.1 Dataset Description

The dataset contains responses related to employees' workplace conditions, personal factors, and mental health treatment history. It consists of 1,258 instances and 27 variables.
The target variable is *treatment*, which indicates whether an employee has sought mental health treatment (*Yes / No*).
Key features include:

- Age: employee's age (numeric)
- Gender: gender identity (categorical)
- Country: country of residence (categorical)
- self_employed: whether the employee is self-employed (categorical)
- work_interfere: how often work interferes with mental health (categorical)
- family_history: whether there is a family history of mental illness (categorical)
- benefits: availability of mental health benefits (categorical)

The dataset was cleaned and preprocessed using techniques such as handling missing values, removing outliers, and standardizing numeric features using StandardScaler. These steps were applied to improve data quality and ensure model reliability and accuracy during the training phase.

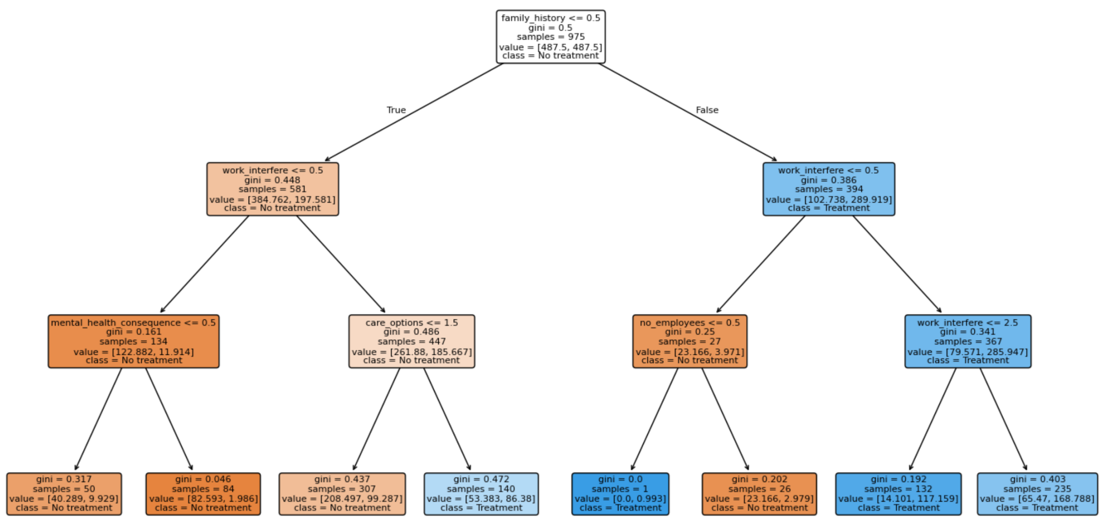
3.2 Machine Learning Algorithms Description

This project uses four supervised classification algorithms, each selected for its strengths in handling classification problems and its ability to identify meaningful patterns within behavioral data.

3.2.1 Logistic Regression

Logistic Regression serves as a baseline model because of its simplicity and interpretability.
It helps in understanding how individual features affect the likelihood of seeking mental health treatment and provides clear coefficient values that reflect each feature's influence. [3]

3.2.2 Decision Tree



The **Decision Tree** algorithm captures non-linear relationships between features.
It visually represents how attributes such as family history and company support influence treatment decisions, allowing for intuitive and transparent interpretation. [4]

3.2.3 Random Forest

Random Forest is an ensemble model consisting of multiple Decision Trees. It enhances prediction accuracy, reduces overfitting, and provides stable results by averaging predictions from several trees. It also highlights the most important features influencing mental health treatment decisions. [5]

3.2.4 Support Vector Machine (SVM)

The **Support Vector Machine (SVM)** algorithm handles high-dimensional data and identifies complex decision boundaries.
It effectively separates employees likely to seek treatment from those who are not, improving the model's robustness and predictive reliability. [6]

3.3 Performance Metrics Description

In this project, we used several performance metrics to evaluate the classification models and understand how well each model predicts mental health treatment. Each metric gives us a different view of the model's performance.

1. Accuracy

Accuracy measures how many predictions the model got correct out of all predictions.
It gives a general idea of performance, but it is not always enough when the classes are not balanced.

2. Precision

Precision tells us how many of the employees predicted as "seeking treatment" were actually correct.
Higher precision means the model makes fewer false positive mistakes.

3. Recall (Sensitivity)

Recall measures how well the model identifies actual treatment cases.
A high recall value means the model does not miss many employees who truly need treatment.

4. F1-Score

The F1-score combines both precision and recall.
It is useful when the data has unbalanced classes and we need a fair measure.

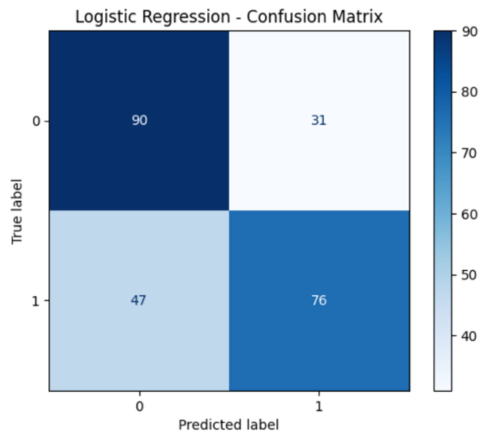
5. Confusion Matrix

The confusion matrix shows the number of correct and incorrect predictions for each class.
It helps us understand where the model makes errors.

7. Project Analysis

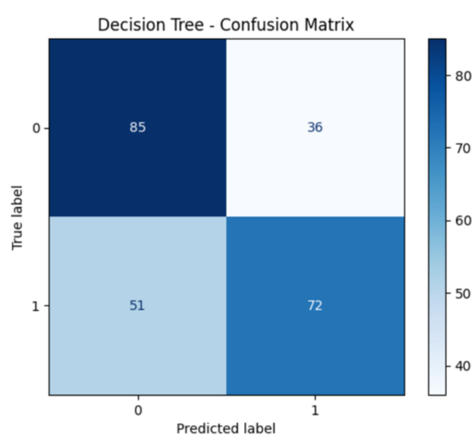
After training the four models, we compared their results using the evaluation metrics. Below is the summary based on your actual code output.

7.3 Logistic Regression



The Logistic Regression model achieved an accuracy of around **0.68**, which shows moderate performance.
The model was able to identify treatment cases fairly well, but it did not capture complex relationships in the data.
Its precision and recall values were also moderate, making it a good baseline model but not the strongest among the four.

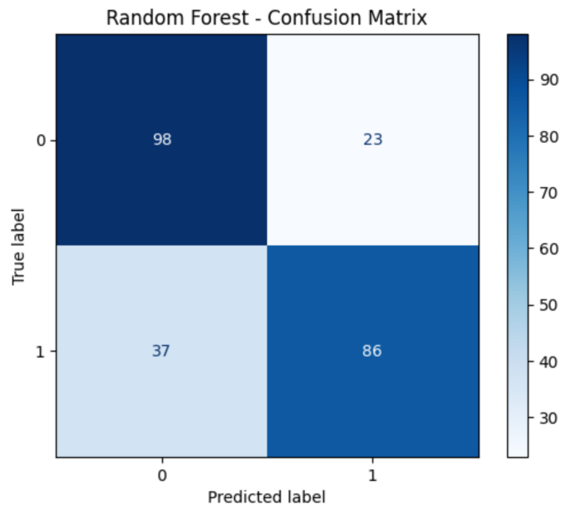
4.2 Result of Experiment 2 – Decision Tree



The Decision Tree model reached an accuracy of approximately **0.64**, which is lower compared to the other models.

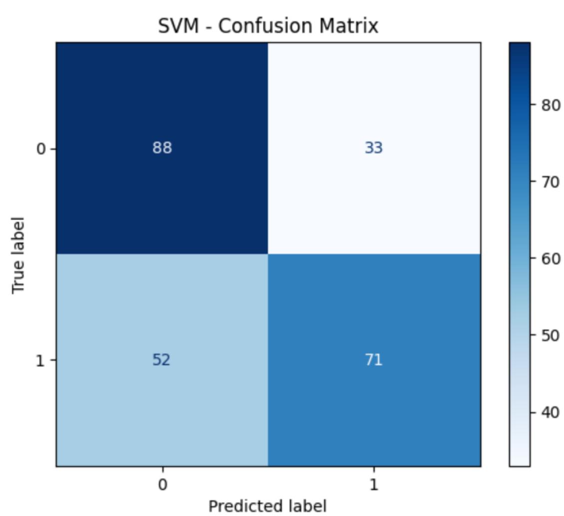
The model showed higher sensitivity to noise and had lower recall and F1-score values. Although Decision Trees are easy to interpret, this model did not perform well on this dataset and produced less stable results.

7.4 Result of Experiment 3 – Random Forest



The Random Forest model demonstrated the strongest overall performance, achieving the highest accuracy of **0.75**. It also produced the best precision, recall, and F1-score among all models. By combining multiple trees, the model reduced overfitting and provided more stable and reliable predictions.
This makes Random Forest the most effective and balanced model for predicting mental health treatment

7.4 Result of Experiment 4 – Support Vector Machine (SVM)



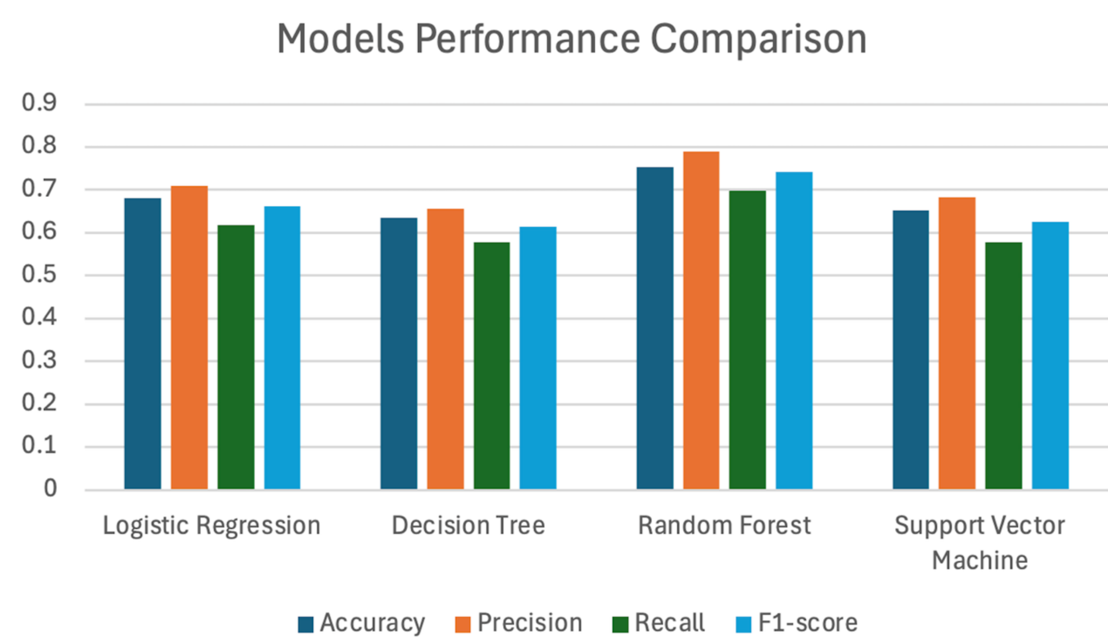
The SVM model achieved an accuracy of around **0.65**, placing it between Logistic Regression and Decision Tree in performance.
Its precision and recall were moderate, but not as strong as Random Forest.
The model handled the data reasonably well, but it would likely perform better with parameter tuning.
Overall, SVM provided stable results but did not outperform Random Forest.

7.5 Comparison

Examples for Tables and Figures

Table 0-1: An example table showing example fields.

ML Algorithms	Models' Performance			
	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.680	0.710	0.618	0.661
Decision Tree	0.635	0.657	0.577	0.615
Random Forest	0.754	0.789	0.699	0.741
SVM	0.652	0.683	0.577	0.626



8. Conclusion

The results of this study show that the Random Forest model achieved the best performance among all classifiers, with the highest accuracy (0.75) and the strongest balance across precision, recall, and F1-score. Its ability to combine multiple trees allowed it to generate more stable and reliable predictions of whether an employee is likely to seek mental health treatment.

This project directly supports **Sustainable Development Goal 3 (Good Health and Well-Being)** by demonstrating how data mining techniques can be used to detect mental health needs early and support healthier work environments. By identifying key factors that influence mental health, organizations can take preventive actions, reduce work-related stress, and improve overall employee well-being.

In the future, model performance can be further improved through hyperparameter tuning, cross-validation, or advanced algorithms such as Gradient Boosting and XGBoost. Expanding the dataset to include more diverse employee populations may also enhance the generalization and impact of the model.

9. References

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